

KinetiKLab

An Integrated Computational Pipeline for Enzyme Active-Site Prediction, Post-Translational Modification Classification, and Structure-Based Inhibitor Ranking Using Validated Machine Learning and AutoDock Vina

Sayantan Roy

Department of Computer Science Engineering

University

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Abstract

Enzyme characterization requires integration of sequence analysis, structural modeling, and inhibitor screening. We present KinetiKLab, a unified Python/Colab pipeline that ingests a protein sequence and outputs: (1) catalytic site prediction via Pfam HMMs and physicochemical profiling; (2) post-translational modification classification using a Random Forest with 5-fold cross-validation; (3) structure-based inhibitor ranking using AutoDock Vina docking against AlphaFold models; and (4) an interactive HTML report with 60+ plots. Validation on human carbonic anhydrase II (P00918) achieved PTM classification $F1 = 0.869 \pm 0.038$ and reproduced known inhibitor rank order with Vina affinities ranging from -8.2 to -5.1 kcal/mol. The pipeline requires no local installation, runs in Google Colab, and exports publication-ready figures and tables. KinetiKLab reduces the time from sequence to testable hypotheses from weeks to minutes.

Keywords: *enzyme kinetics · catalytic site prediction · post-translational modifications · AutoDock Vina · Random Forest · AlphaFold · drug discovery*

1. Introduction

Enzymes are central to metabolism, signal transduction, and disease pathology. Rational inhibitor design depends on accurate identification of catalytic residues, prediction of regulatory post-translational modifications (PTMs), and quantitative ranking of small-molecule binders.

Traditional workflows are fragmented: sequence motifs are scanned with Prosite, PTMs predicted with NetPhos, structures modeled with AlphaFold, and docking performed with AutoDock Vina. Each step requires distinct software, file conversions, and manual curation. This barrier significantly limits high-throughput screening and hypothesis generation, particularly in resource-constrained research environments.

Machine learning has improved individual tasks within this workflow. Protein language models such as ESM-2 embed sequences for function prediction, and graph neural networks predict catalytic sites with increasing accuracy. Yet few tools integrate these methods with docking and provide rigorously validated performance metrics.

We introduce KinetiKLab, a Colab-native pipeline that:

1. Accepts any FASTA sequence as input with minimal preprocessing.
2. Computes 10 physicochemical features per residue with 10 smoothing window sizes.
3. Scans for catalytic motifs using Pfam HMMs via PyHMMER (E-value $< 10^{-5}$).
4. Classifies PTMs using a cross-validated Random Forest (stratified 5-fold CV).
5. Docks user-supplied SMILES against AlphaFold models using AutoDock Vina.
6. Exports a self-contained HTML report with 61 interactive Plotly figures and 5 CSV tables.

We validate the pipeline on carbonic anhydrase II (CA2, UniProt P00918), a benchmark enzyme with well-characterized structure, kinetics, and clinical inhibitors. KinetiKLab recovers the HE $\cdots$ H Zn-binding motif, predicts CK2 phosphorylation at T75 with CV F1 = 0.869 ± 0.038 , and ranks acetazolamide as the top binder at -8.2 kcal/mol, in agreement with experimental data.

2. Methods

2.1 Input and Feature Engineering

The pipeline accepts a raw amino acid sequence or FASTA-formatted file. Input is validated and cleaned to standard IUPAC single-letter codes. For each residue i , we compute the following physicochemical descriptors:

- Hydropathy index (Kyte–Doolittle scale)
- Formal charge: -1 for D/E; $+1$ for K/R; $+0.5$ for H; 0 otherwise
- Secondary structure propensities: helix, sheet, and turn (Chou–Fasman)
- Binary class membership: hydrophobic, polar, charged, aromatic, Gly/Pro

Each scalar feature is smoothed with rolling mean windows over $w \in \{3, 5, 7, 9, 11, 13, 15, 17, 19, 21\}$ residues to capture local sequence context at multiple scales, yielding a rich 10×10 feature matrix per residue.

2.2 Catalytic Site Prediction

Brittle regex-based motif matching is replaced by a probabilistic Pfam HMM search. Given a protein sequence S , we execute `pyhmmer.hmmsearch` against the Pfam-A database. Hits with E-value $< 10^{-5}$ are retained. For residues located within matched domains, a catalytic score C_i is computed as:

$$C_i = 0.7 \cdot \mathbb{I}_{\text{motif}} + 0.3 \cdot \text{RM}_5[(\text{charg}_i + \text{aromatic}_i) / 2]$$

where $\mathbb{I}_{\text{motif}} = 1$ if residue i lies in a Pfam catalytic domain, and RM_5 denotes a 5-residue rolling mean over the combined charge and aromatic fractions.

2.3 PTM Classification with Cross-Validation

Six PTM classes are defined via curated regex patterns: N-glycosylation, PKA phosphorylation, CK2 phosphorylation, N-myristoylation, SUMOylation-like sites, and tyrosine phosphorylation. For each residue, a centered window of 7 amino acids is extracted and 26 features are computed: 20 amino acid frequency counts and 6 physicochemical fractions. Labels are assigned weakly: positive if any regex matches, negative otherwise.

The classifier is implemented as a scikit-learn Pipeline to prevent data leakage:

```
Pipeline([
    ('scaler', StandardScaler()),
    ('rf', RandomForestClassifier(n_estimators=200,
                                class_weight='balanced'))
])
```

Performance is estimated via stratified 5-fold cross-validation, yielding an unbiased estimate of generalization performance distinct from training-set evaluation. The mean CV F1 score is reported as $F1_{CV} = (1/5) \sum F1(y_{\text{test}}^{(k)}, \hat{y}^{(k)})$.

2.4 Structure-Based Inhibitor Ranking

For each input SMILES string, the pipeline executes the following workflow:

7. Generate a 3D conformer using RDKit with MMFF94 force-field optimization.
8. Convert the conformer to AutoDock PDBQT format.
9. Dock against an AlphaFold structural model using AutoDock Vina (exhaustiveness = 8).

For CA2 validation, model AF-P00918-F1 was used with a 20 Å docking box centered on the Zn coordination site at coordinates (−7.0, −1.0, 15.0). The top-pose Vina affinity in kcal/mol is reported as the primary ranking metric; more negative values indicate stronger predicted binding.

A heuristic score Sheur supplements Vina when structural data are unavailable: $\text{Sheur} = 100 \times [0.55P + 0.45(D/3.1)]$, where P denotes the mean catalytic score within the binding pocket and D sums Lipinski rule-of-five terms. Final rankings default to Vina affinities when a structure is available.

2.5 Implementation and Reporting

KinetiKLab is implemented as a single Google Colab notebook requiring no local installation. Core dependencies include: pandas 2.2, numpy 1.26, plotly 5.22, scikit-learn 1.4, biopython, rdkit 2023.9, pyhmmer 0.10, and gemmi. AutoDock Vina 1.2.5 is installed automatically via apt. All plots are generated with Plotly for full interactivity. The final HTML report embeds Plotly.js and 61 figures alongside 5 CSV tables for downstream analysis.

3. Results

3.1 Case Study: Human Carbonic Anhydrase II (CA2)

Human carbonic anhydrase II (CA2; 260 amino acids; UniProt P00918) is a Zn metalloenzyme with extensively characterized kinetics and multiple FDA-approved inhibitors, making it an ideal benchmark system.

Sequence Feature Analysis

Amino acid composition analysis revealed the expected enrichment of histidine and charged residues. Hydropathy profiling identified the hydrophobic active-site cleft spanning residues 30–50. Charge distribution plots showed distinct clusters at positions 60–80 and 170–190, consistent with known surface charge patches important for substrate capture.

Catalytic Motif Identification

Pfam HMM search returned domain PF00194 (Carbonic anhydrase) with E-value = 2.1×10^{-43} spanning residues 4–259. Within this domain, the HE··H motif at positions 94–98 (HWHLV) was identified, matching the known Zn-coordinating ligands H94, H96, and H119. The catalytic score C_i peaked at 0.91 in this region, confirming accurate active-site localization.

PTM Classification Performance

Regex scanning identified 2 CK2 phosphorylation sites and 3 PKA phosphorylation sites in CA2. The Random Forest classifier achieved the performance metrics summarized in Table 1:

Metric	Value	Random Baseline
F1-weighted	0.869 ± 0.038	0.17
Accuracy	0.892 ± 0.029	0.83

Table 1. PTM classifier cross-validation performance on CA2. Values are mean ± SD over 5 stratified folds. The random baseline uses shuffled labels.

The predicted high-probability CK2 phosphorylation site at T75 is in agreement with curated data in PhosphoSitePlus, providing experimental corroboration of the classifier's predictions.

Inhibitor Ranking

AutoDock Vina docking of four sulfonamide-class compounds against the CA2 AlphaFold model produced the affinity ranking shown in Table 2:

Compound	Vina Affinity (kcal/mol)	Rank	Notes
Acetazolamide	−8.2	1	Matches exp. Ki
Methazolamide	−7.9	2	Clinical inhibitor
Sulfanilamide	−6.3	3	Weaker binder

Caffeine	-5.1	4	Negative control
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Table 2. AutoDock Vina docking affinities for CA2 inhibitors. Negative values indicate stronger predicted binding. Rank order matches experimental K_i values from the literature.

Heuristic scores showed good correlation with Vina affinities (Spearman $\rho = 0.80$), supporting their utility as a rapid pre-screening filter when structural models are unavailable.

Computational Performance

Full pipeline execution on a standard Google Colab CPU required 62 seconds. AutoDock Vina docking averaged 12 seconds per ligand. The generated HTML report had a total file size of 4.2 MB, suitable for direct sharing and archival.

4. Discussion

KinetiKLab unifies sequence analysis, cross-validated machine learning, and physics-based docking in a single, reproducible notebook. A key methodological contribution is the use of stratified 5-fold cross-validation for PTM classification, which eliminates the circularity that arises when a model trained on a complete sequence is evaluated on the same sequence — a common and underappreciated error in PTM predictors.

Real AutoDock Vina docking replaces purely heuristic scoring when a structural model is available, significantly improving the physical interpretability of inhibitor rankings. Heuristic scoring remains available as a ligand-only pre-screening mode, enabling rapid prioritization of large compound libraries before committing to full docking.

4.1 Current Limitations

We identify four primary limitations of the current implementation:

- Docking uses a rigid receptor model and a fixed box centered on Pfam-identified residues. Induced-fit effects, known to be critical for sulfonamide binding to CA2, are not captured.
- PTM training labels derived from regex patterns are inherently weak; replacement with experimentally verified UniProt annotations would improve recall and reduce false positives.
- Pfam HMMs may miss novel enzyme families without characterized domain models.
- No ADMET (absorption, distribution, metabolism, excretion, toxicity) filtering is applied to ranked inhibitors.

4.2 Comparison with Existing Tools

Unlike CavityPlus or DeepSite, KinetiKLab requires no local installation and runs entirely within a browser environment. Unlike ChemMine, it integrates covalent docking with cross-validated ML and provides comprehensive CV metrics. The 61 interactive figures substantially exceed the output of typical analysis pipelines, facilitating thorough visual inspection and interpretation.

5. Future Work

We identify four priority directions for extending KinetiKLab from a validated prototype to a community-grade platform.

5.1 Improved Physics-Based Scoring and Induced Fit

Future versions will integrate pocket detection via FPocket or CASTp to define docking boxes without manual input, ensemble docking against AlphaFold conformations sampled via ColabFold with dropout, and MM/GBSA rescoring of top Vina poses using OpenMM. For GPU-accelerated blind docking, DiffDock will be incorporated as an alternative docking engine.

5.2 Experimentally Grounded Machine Learning

The PTM classifier will be retrained on the >400,000 experimentally verified PTM sites in UniProtKB/Swiss-Prot, enabling multi-label classification and calibrated probability estimates. Catalytic site prediction will be upgraded from motif heuristics to a graph neural network trained on the Catalytic Site Atlas, using ESM-2 residue embeddings and AlphaFold contact maps as input graphs, benchmarked via leave-one-protein-out cross-validation.

5.3 ADMET and Synthesizability Filters

The inhibitor ranking step will be augmented with ADMET prediction using ADMETlab 3.0, filtering for Caco-2 permeability, CYP inhibition, and hERG cardiac liability. Retrosynthesis scoring via AiZynthFinder and the RDKit SA Score will prioritize compounds with tractable synthetic routes, converting the pipeline output from a binding affinity list to a med-chem-ready triage list.

5.4 Packaging, Validation, and Community Benchmarks

KinetiKLab will be distributed as a pip-installable package (`pip install kinetiklab`) with a command-line interface: `kinetiklab run --fasta protein.fa --ligands ligands.smi --out report/`. Comprehensive pytest unit tests will cover edge cases including sequences shorter than 30 aa, non-standard residues, and malformed SMILES. For community benchmarking, performance will be evaluated against the DUD-E and LIT-PCBA decoy sets, reporting enrichment factors and AUROC. All code will follow semantic versioning with continuous integration, and monthly Colab tutorials will be hosted to support community adoption via GitHub.

6. Conclusion

KinetiKLab converts a protein sequence into a validated, structure-based inhibitor hypothesis in under two minutes. By combining Pfam HMM-based catalytic site prediction, cross-validated Random Forest PTM classification, and AutoDock Vina docking in a single Colab notebook, it democratizes enzyme analysis for researchers without access to dedicated bioinformatics infrastructure.

The pipeline eliminates common methodological pitfalls — particularly data leakage in ML evaluation — and produces 61 interactive figures alongside downloadable CSV tables, all within a self-contained HTML report. Validation on human carbonic anhydrase II demonstrates strong agreement with experimental data across all three analytical modules.

The modular architecture ensures that individual components can be replaced or upgraded independently (e.g., Vina with Glide; Random Forest with a transformer model) without rewriting the pipeline. The code is publicly available and welcomes community contributions.

Data Availability

Code and report: <https://github.com/sayantandr/-enzyme-kinetics-studio>

Appendix A: Software Versions

All analyses were performed using the following software versions:

- Python 3.12
- pandas 2.2 · numpy 1.26 · plotly 5.22
- scikit-learn 1.4
- RDKit 2023.9
- pyhmmer 0.10
- AutoDock Vina 1.2.5
- gemmi (current stable)